

RESEARCH ARTICLE

A Delphi Study on Ethical Challenges and Ensuring Academic Integrity Regarding AI Research in Higher Education

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ABSTRACT

The rapid integration of artificial intelligence (AI) into higher education has revolutionised academic research and teaching, offered transformative opportunities while raising significant ethical challenges. This Delphi study investigates the ethical dilemmas and institutional requirements for maintaining academic integrity in AI-driven research. Through a structured, iterative process involving 12 international experts from academia, AI regulatory bodies, and research ethics committees, the study identifies key concerns such as data privacy, algorithmic bias, transparency, and authorship in AI-assisted research. The findings reveal a pressing need for clear ethical guidelines, robust institutional oversight, and enhanced collaboration between academia and industry to ensure responsible AI use. Experts emphasise the importance of developing enforceable policies, integrating AI ethics into academic curricula, and fostering transparency in AI decision-making processes. The study concludes that while AI offers immense potential for innovation, its ethical implications demand proactive governance and continuous policy refinement. By addressing these challenges, higher education institutions can uphold academic integrity, mitigate risks of misconduct, and promote the responsible development and application of AI technologies. This research provides actionable recommendations for policymakers, educators, and researchers, contributing to the creation of a sustainable and ethically sound AI ecosystem in academia.

1 | Introduction

The integration of artificial intelligence (AI) into higher education and research has reshaped academic practices, from data analysis to content creation. While AI offers numerous benefits, it also raises significant ethical concerns, particularly regarding academic integrity and research ethics. The ability of AI tools to generate human-like text has sparked debates on plagiarism, authorship, and the credibility of scholarly work (Currie 2023; Liman-Kaban 2025). These challenges necessitate clear institutional policies to ensure responsible AI use in academia. Despite the increasing presence of AI in academia, many institutions have yet to establish comprehensive policies that effectively

address these ethical dilemmas. The absence of clear guidelines on the responsible use of AI in academic and research settings poses risks to the quality and credibility of scholarly outputs, potentially undermining public trust in academic research. To preserve academic excellence and integrity, it is essential to bridge this gap and develop well-defined institutional frameworks.

For decades, scientific malpractices—ranging from questionable research practices to severe misconduct—have been recognised as significant issues that produce misleading results, waste public funds, erode trust in the research process, increase regulatory oversight, and hinder scientific progress (Ioannidis et al. 2014). In response, strengthening ethics and research

integrity (ERI) has become a critical component of science education and training. However, the effectiveness of ERI training remains a subject of debate, as studies have produced mixed results. While some research has found no significant impact of ERI education on ethical behaviour (Antes et al. 2010; May and Luth 2013), other studies have demonstrated substantial benefits and improvements in training methods in recent years (Watts et al. 2017).

Two primary approaches dominate ERI training: a principle-based approach, which emphasises adherence to ethical rules, duties, and responsibilities (Resnik 2012), and a virtue-based approach, which focuses on character development and situational ethical reasoning, enabling scientists to strive for excellence in their practices (Chen 2015; Pennock and O'Rourke 2017). Critics of the principle-based approach argue that it often fails to provide adequate guidance for ethical decision-making in real-life situations, as it focuses narrowly on compliance and neglects the development of intrinsic values (Steele et al. 2016). While some theorists favour virtues over principles (Macfarlane 2009), most agree that ERI training should integrate both approaches, as they complement each other—one emphasising rule-following and the other character development (Resnik 2012).

While much of the existing literature on science and virtue ethics has focused on moral-philosophical aspects, there is a growing interest in its educational applications (Chen 2015; Horbach and Halffman 2017). However, there is no consensus on which virtues are most important for research integrity, as various scholars and academic organisations have proposed different lists of essential scientific virtues (ALLEA 2017; Pennock and Miller 2017; Liman-Kaban 2025). This lack of agreement poses practical challenges for developing standardised training materials and curricula. To address this gap, this study aims to engage a broad range of ERI experts to build consensus on the most important virtues to prioritise in training for good research practices and how they should be effectively taught. By doing so, it seeks to advance the evidence base for virtue-based ERI training and contribute to more effective and holistic approaches to fostering research integrity.

This study explores the institutional needs and challenges associated with maintaining academic integrity and research ethics in the era of AI. By utilising the Delphi method with a panel of experts, the research aims to establish a consensus on effective strategies and policies that institutions can implement. The goal is to provide actionable recommendations that will help academic institutions navigate the evolving ethical landscape shaped by AI technologies.

2 | Literature Review

The incorporation of AI into academic settings has fuelled widespread discussions about its impact on academic integrity. AI serves a dual purpose—enhancing educational experiences while also posing risks related to academic misconduct—making it essential to approach its ethical and practical implications with a balanced perspective (Currie 2023). As AI-generated content grows more advanced, conventional methods for detecting plagiarism and evaluating academic work encounter increasing

difficulties (Eaton 2022). This review delves into the intersection of AI and academic integrity by analysing key concerns, evolving ethical frameworks, and institutional policy responses within higher education.

A major concern surrounding the use of AI in academic environments is its potential role in enabling academic misconduct. With the emergence of generative AI tools capable of producing sophisticated text, the likelihood of AI-assisted plagiarism has increased considerably (Eaton 2022; Liman-Kaban 2025). Conventional plagiarism detection systems often struggle to distinguish between content written by humans and that generated by AI, making existing academic integrity policies less effective (Chan 2023).

Additionally, the issue of authorship in AI-assisted research presents significant ethical challenges. Questions regarding intellectual ownership—whether attribution should go to the human researcher, the AI system, or both—continue to spark debate. Under the 1976 Copyright Law, AI-generated works are ineligible for copyright protection, as they require human authorship (Schroeder 2023). The debate on whether AI can be credited as a co-author is ongoing. Some argue that AI should not be credited as a co-author due to its lack of human reasoning and accountability (Bakla 2023; Bozkurt 2024; Teixeira da Silva and Tsigaris 2023). Lin (2024) emphasises the importance of establishing clear ethical guidelines to define authorship conventions and promote transparency in academic work involving AI. These challenges highlight the urgent need for a structured ethical framework to regulate AI's role in scholarly contexts.

To address these challenges, institutions have begun formulating ethical frameworks, and policy frameworks play a crucial role in guiding the ethical use of AI in academic settings. The Ethical Principles AI Framework for Higher Education provides a foundational structure for ensuring responsible AI implementation, emphasising institutional adaptability and compliance with evolving technological advancements (Wynants et al. 2025). In a similar vein, Chen et al. (2024) explore AI's broader impact on research integrity, advocating for well-defined ethical policies to mitigate potential risks associated with AI-generated academic outputs.

The *Delphi method* has emerged as a valuable approach for reaching expert consensus on ethical challenges related to AI. This structured process fosters informed discussions among specialists, facilitating the identification of critical risks and the development of strategic policy responses (Bannister et al. 2023). Hsu and Sandford (2007) highlight how the iterative nature of the Delphi technique supports the creation of well-founded and collectively agreed-upon academic integrity guidelines. By utilising expert-driven insights, institutions can take proactive measures to address AI-related ethical concerns before they escalate into widespread issues.

Universities are actively addressing the growing presence of generative AI by establishing comprehensive policies to regulate its use. Dabis and Csáki (2024) examine how higher education institutions have initially responded to this challenge, revealing a range of regulatory approaches. While some universities have enforced strict limitations on AI-generated content in academic

work, others have prioritised educating students and faculty on ethical AI usage.

Chan (2023) introduces a structured AI policy education framework aimed at embedding AI literacy within university curricula. By ensuring that both students and educators develop a strong understanding of AI's ethical and practical implications, institutions can encourage responsible usage while safeguarding academic integrity. Furthermore, Currie (2023) emphasises the need to strike a balance—leveraging AI's educational benefits while implementing measures to deter academic misconduct.

Despite ongoing efforts, the rapid advancements in AI technology continue to outstrip the development of ethical guidelines and institutional regulations. This gap creates a constantly shifting landscape where maintaining academic integrity remains a persistent challenge (Chen et al. 2024). To address this issue, institutions must take a proactive approach by regularly updating policies and engaging in interdisciplinary research to anticipate emerging concerns.

One effective strategy involves incorporating AI ethics into academic curricula to instil a sense of responsibility among both students and educators (Dabis and Csáki 2024). Additionally, fostering collaboration between universities, policymakers, and AI developers can contribute to the creation of stronger ethical frameworks that comprehensively address AI's implications in academia. Complementing these institutional strategies, Radanliev (2025) proposes an ethical AI development framework that aligns with global policy variations from the EU, USA and China. His work identifies transparency, fairness, and privacy as intersecting pillars for responsible AI design and offers actionable steps, such as routine audits, fairness-aware algorithms and diverse development teams, to promote ethical compliance across educational AI ecosystems.

Navigating the intersection of AI and academic integrity requires ongoing assessment and adaptation. While AI offers significant educational benefits, it also presents ethical and procedural challenges that demand well-defined policies and regulatory frameworks. By utilising structured methodologies such as the Delphi technique and formulating comprehensive AI policies, higher education institutions can uphold academic integrity in an increasingly digital environment. Future research should focus on refining ethical standards and exploring innovative assessment methods that align with AI's evolving capabilities. Ensuring academic integrity in the AI-driven era will require sustained collaboration among educators, policymakers, and technology experts to promote ethical and responsible AI use in academia.

3 | Methodology

This study employs the Delphi method, a structured and iterative research approach designed to systematically gather expert insights and facilitate consensus on complex and multifaceted issues (Hsu and Sandford 2007). Widely utilised in fields such as forecasting, decision-making, and expert-driven studies, the Delphi technique is particularly advantageous in domains

characterised by uncertainty, rapid technological evolution, and the need for informed judgement (Dalkey and Helmer 1963; Rowe and Wright 1999). Given the dynamic and interdisciplinary nature of AI ethics, the Delphi method is especially well suited for this research, as it enables the identification of critical ethical challenges and the development of actionable strategies to uphold academic integrity in AI-related research. By engaging a diverse panel of experts in iterative rounds of discussion and feedback, this approach ensures a robust and comprehensive exploration of the ethical dimensions of AI in academia, ultimately fostering a consensus-driven framework for addressing these challenges.

4 | Research Design

This study employs a two-round Delphi method, involving a diverse panel of international experts specialising in AI research, academic integrity, and ethics. The iterative nature of this approach allows for the systematic refinement of expert insights, minimising individual biases while fostering collective reasoning (Rowe and Wright 1999). By maintaining a structured process, the Delphi method effectively identifies critical themes, encourages meaningful discourse, and guides the panel toward a consensus, making it particularly well suited for examining ethical dilemmas in AI research (Dalkey and Helmer 1963; Turoff and Linstone 2002).

Although traditional Delphi studies often extend to three or more rounds, this study concludes after two rounds based on established best practices and practical considerations. Prior research indicates that when a strong consensus emerges within the first two rounds, additional iterations tend to produce diminishing returns, adding redundancy rather than new insights (Okoli and Pawlowski 2004; Skulmoski et al. 2007; Roth et al. 2017). In this case, the second round demonstrated a significant alignment of expert perspectives, confirming that further rounds would not provide substantial new contributions or alter the study's key conclusions (Hsu and Sandford 2007; Rowe and Wright 1999).

5 | Expert Selection and Sampling

The selection of qualified experts constitutes a foundational element of the Delphi methodology, as the validity, reliability, and overall robustness of the study's findings are intrinsically linked to the depth of expertise and diversity of perspectives among participants. Unlike methodologies that prioritise statistical representativeness, the Delphi approach is predicated on informed judgement, leveraging the collective insights of individuals with substantial knowledge and experience in the subject area (Rowe and Wright 1999; Okoli and Pawlowski 2004). Consequently, purposive sampling emerges as a strategically appropriate method for identifying and engaging participants who possess specialised expertise, ensuring that the study benefits from a rich and multifaceted examination of the topic. This approach not only enhances the credibility of the findings but also facilitates the development of nuanced, consensus-driven recommendations that reflect

the informed assessments of a well-versed and diverse expert panel. The selected experts represent a broad spectrum of professional backgrounds, allowing for a multidimensional examination of the ethical complexities introduced by AI in academic environments. The participant pool includes individuals from the following domains:

- *Academia*: AI researchers, ethicists, and scholars specialising in academic integrity, with established publication records and contributions to discussions on responsible AI usage.
- *AI Regulatory Bodies*: Policymakers and legal professionals engaged in the development and enforcement of AI governance frameworks, ensuring alignment with evolving regulatory standards.
- *Research Ethics Committees*: Specialists responsible for evaluating ethical compliance in AI-related studies, particularly in higher education and scientific research settings.

By incorporating *experts from diverse sectors*, this Delphi study ensures a well-rounded analysis, mitigating potential biases that may arise from a homogeneous expert panel (Skulmoski et al. 2007). The inclusion of individuals with practical policy-making experience, combined with those engaged in theoretical and empirical research, provides a *comprehensive perspective* on the ethical dilemmas posed by AI in academic environments. An overview of the participants' areas of expertise and institutional affiliations is provided in Table 1, and this table aims to highlight the interdisciplinary and cross-institutional nature of the expert panel.

A total of 12 experts were invited to participate, all of whom completed the study across multiple Delphi rounds, ensuring a 100% retention rate. The absence of participant dropout is a notable strength, as it enhances the continuity, reliability and robustness of the consensus-building process. This consistent engagement throughout the Delphi rounds contributed to the refinement of expert opinions and the development of nuanced policy recommendations.

6 | Selection Criteria

Participants were selected based on a rigorous set of inclusion criteria designed to ensure that only individuals with significant expertise and contributions in the field were involved. The criteria for expert selection included:

1. Recognised expertise in AI ethics, academic integrity, or higher education governance, demonstrated through professional roles, research output or policy engagement.
2. A strong publication record in peer-reviewed journals, or active contributions to policy development and institutional governance related to AI in academia.
3. Extensive professional experience in AI research, research ethics committees or institutional policymaking related to academic integrity and AI regulations.

TABLE 1 | Participant expertise and affiliation table.

Participant	Area of expertise	Institutional affiliation
Participant 1	Educational Technology	State University
Participant 2	Educational Technology	State University
Participant 3	Artificial Intelligence Consultancy	AI Consultancy Company
Participant 4	Artificial Intelligence Research	State University
Participant 5	Artificial Intelligence Research/Teaching	Private School
Participant 6	Educational Technology	Private University
Participant 7	Artificial Intelligence Enthusiast/Teaching	Ministry of National Education (Turkey)
Participant 8	Educational Technology	Private University
Participant 9	Artificial Intelligence Research	CERN
Participant 10	Educational Technology—HE Turkey Board Member	Private University
Participant 11	E-learning	State University
Participant 12	Educational Technology—Entrepreneurship—Ethics Committee Member of a Private College	Private Company

4. Willingness to engage in multiple Delphi rounds, providing iterative feedback and contributing to consensus-building through structured deliberation.

The Delphi methodology relies on expert judgement rather than broad statistical sampling (Rowe and Wright 1999; Okoli and Pawlowski 2004), making it imperative to select participants who possess a deep understanding of AI-related ethical issues and are capable of evaluating complex dilemmas within higher education. The criteria ensured that each participant brought a unique and informed perspective, enriching the collective analysis and reinforcing the study's credibility.

7 | Delphi Process and Expert Engagement

The study was conducted using a two-round Delphi process, allowing for structured refinement of expert perspectives through iterative feedback. In the first round, experts were asked to

provide open-ended responses to key ethical questions concerning AI's impact on academic integrity, challenges in policy implementation, and the broader implications of AI in scholarly settings. These qualitative insights were systematically analysed and synthesised into thematic categories, forming the basis for the second round.

The second Delphi round involved structured questions designed to assess consensus on the key themes identified in the initial phase. Participants were asked to rate statements using a 5-point Likert scale, with additional opportunities to provide qualitative justifications for their responses. This iterative process facilitated deeper engagement with the subject matter, refining expert opinions while maintaining a collaborative and rigorous analytical approach.

The absence of participant dropout across both rounds is particularly significant, as Delphi studies often face attrition challenges due to the demanding nature of repeated expert engagement. The full retention of all 12 participants strengthens the credibility and stability of the findings, ensuring that the conclusions drawn from the study are based on consistent and fully developed expert insights.

7.1 | Round 1—Open-Ended Questions

The first phase involves participants responding to open-ended questions regarding the ethical dimensions of AI in academic research. Key topics explored include:

- Institutional policies on AI-generated content and academic integrity.
- Challenges in identifying and addressing AI-assisted academic misconduct.
- Ethical considerations in utilising AI for research and assessment purposes.
- Strategies for ensuring integrity in AI-driven academic work.

This stage facilitates the collection of diverse viewpoints and helps identify emerging themes within the discourse (Güneş 2022).

7.1.1 | Sample Open-Ended Questions (Round 1)

In the first round of the Delphi process, participants were asked to respond to the following open-ended questions:

1. What do you consider to be the most critical ethical challenges encountered in AI research in higher education? Can you provide examples to illustrate your views?
2. How would you describe the institutional barriers to ensuring academic integrity in AI research? What steps do you think can be taken to overcome these obstacles?
3. What are your thoughts on the strengths and weaknesses of the current AI policies? In which areas do you believe improvements are most urgently needed?

4. Which fundamental principles should be considered when establishing ethical standards in AI research?
5. In your opinion, how can collaboration between academic institutions and the AI sector be strengthened to support the adoption and enforcement of ethical standards in AI research?

7.2 | Round 2—Refinement and Thematic Analysis

Findings from the initial round are synthesised into key themes, and participants are then asked to evaluate these emerging ethical concerns and strategies using a Likert scale. This stage enables:

- Prioritisation of significant ethical challenges through quantitative assessment.
- Identification of patterns and areas where expert perspectives diverge.
- A structured synthesis of qualitative insights to ensure clarity and consensus.

Results from the second round revealed a strong alignment in expert viewpoints, indicating that further iterations would not significantly alter the study's conclusions. Research has demonstrated that when consensus is largely achieved by the second round, additional rounds often lead to diminishing returns without contributing meaningful new insights (Okoli and Pawlowski 2004; Skulmoski et al. 2007). Consequently, the decision was made to finalise the Delphi process after the second round.

This study employs a rigorous two-round Delphi approach, engaging AI and ethics experts worldwide to build consensus on AI's role in academic integrity. By incorporating structured feedback loops, maintaining participant anonymity, and iteratively refining responses, the study ensures that the insights gathered are valid, reliable, and comprehensive. Concluding the study after two rounds aligns with best practices in Delphi research, upholding methodological rigour while ensuring efficiency and avoiding unnecessary redundancy in expert input.

8 | Findings and Discussion

8.1 | Analysis of Round One Expert Responses

This section provides a comprehensive examination of the insights shared by experts in the first round of the Delphi study. The responses highlight critical ethical challenges, institutional barriers, policy effectiveness, ethical standards and collaboration between higher education institutions and the AI sector. The analysis integrates direct participant perspectives while identifying key themes requiring further discussion and refinement.

8.1.1 | Ethical Challenges in AI Research in Higher Education

Experts identified multiple ethical concerns in AI research, emphasising the need for stricter regulations and increased transparency:

- *Handling of Personal Data and Privacy Issues:* Several participants highlighted concerns over how personal data is accessed and used in AI research. Participant 2 stressed that ‘personal information should be protected by strict regulations to prevent unauthorized use and potential privacy violations.’ Participant 1 echoed similar concerns, noting the ‘lack of clear ethical responsibility among researchers, making compliance with fair data practices uncertain.’ These responses indicate that current safeguards are either insufficient or inconsistently enforced.
- *Bias in AI Algorithms and Fairness:* The risk of bias in AI decision-making emerged as a major ethical concern. Participant 3 warned that ‘AI models trained on biased datasets may disproportionately disadvantage specific student groups, perpetuating existing inequalities rather than mitigating them.’ This perspective underscores the need for fairness assessments in AI-driven academic tools.
- *Lack of Transparency in AI-Generated Decisions:* The complexity of AI decision-making raises accountability concerns. Participant 4 stated that ‘without clear documentation on how AI models arrive at their conclusions, ensuring ethical oversight becomes nearly impossible.’ Experts emphasised that AI-generated outputs must be interpretable and justifiable to maintain trust and academic integrity.

These findings suggest that strengthening ethical regulations, implementing bias mitigation strategies, and ensuring transparency in AI processes should be prioritised in academic institutions.

8.1.2 | Institutional Barriers to Upholding Academic Integrity in AI Research

Participants identified multiple structural challenges that hinder the enforcement of ethical AI practices:

- *Inadequate Institutional Oversight:* Participant 1 expressed concern that ‘academic integrity violations, particularly in AI-generated research, often go unchecked, leading to the normalization of unethical practices.’ This highlights a need for more stringent monitoring mechanisms within institutions.
- *Lack of Transparency and Accountability:* Participant 3 argued that ‘opaque AI systems, when deployed in academic settings without clear oversight, create loopholes that can be exploited, undermining academic trust.’ Ensuring greater transparency in AI research methodologies was proposed as a solution.
- *Institutional Focus on Productivity Over Ethics:* Participant 2 observed that ‘universities prioritize efficiency and research output, often sidelining ethical considerations in the rush to produce AI-driven academic tools.’ This suggests that institutional priorities need rebalancing to integrate ethical responsibility alongside innovation.

Participants recommended developing stronger governance mechanisms to address these institutional weaknesses and uphold ethical AI research standards.

8.1.3 | Evaluating the Strengths and Weaknesses of Current AI Policies

Experts assessed existing AI regulations, identifying both their benefits and shortcomings:

- *Strengths:*
 - Participant 2 acknowledged that ‘current AI policies set basic ethical boundaries, ensuring at least a foundational level of responsible AI usage.’
 - Participant 3 noted that ‘AI regulations are evolving to keep pace with technological advances, but often reactively rather than proactively.’
- *Weaknesses:*
 - *Participant 4* raised concerns that ‘many AI ethics policies are theoretical rather than enforceable, limiting their effectiveness in practice.’
 - *Participant 1* criticised the policies for ‘being too general and lacking specific, actionable guidelines, leaving institutions without clear direction.’

These insights suggest that AI governance frameworks must transition from broad principles to concrete, enforceable regulations to ensure real-world applicability.

8.1.4 | Key Principles for Ethical AI Research

Experts emphasised several core principles that should guide ethical AI development and deployment:

- *Protecting Individual Rights and Privacy:* Participant 1 highlighted that ‘AI frameworks should prioritize safeguarding personal data and ensuring individuals’ rights to privacy are upheld at all times.’
- *Ensuring Transparency and Accountability:* Participant 3 suggested adopting ‘clear accountability mechanisms, such as those outlined in the EU AI Act, to maintain ethical AI oversight.’
- *Defining AI Use Cases and Boundaries:* Participant 4 stressed that ‘AI applications should have well-defined limits, ensuring that their roles and interventions are transparent to all stakeholders.’

These principles reinforce the necessity of structuring AI ethics policies around transparency, accountability, and individual rights.

8.1.5 | Strengthening Academic and Industry Collaboration on AI Research

Participants proposed various strategies to improve ethical partnerships between universities and AI companies:

- *Highlighting Successful Ethical AI Implementations:* Participant 1 suggested that ‘showcasing real-world examples of ethical AI use in academia can inspire institutions to adopt best practices.’

- Enhancing University-Industry Collaborations: Participant 2 noted that ‘AI research partnerships between universities and the private sector remain weak, requiring structured collaboration models to ensure ethical standards are upheld.’
- Prioritising Ethics in AI Research Funding: Participant 3 recommended that ‘funding organizations should prioritize AI projects that integrate strong ethical safeguards and align with societal well-being.’

These recommendations indicate that ethical considerations should be central to AI-academic partnerships rather than being an afterthought.

The first round of the Delphi study has provided a critical foundation for understanding expert perspectives on AI ethics in academia. The findings highlight the urgent need for clearer ethical guidelines, stronger institutional oversight, and refined AI policy frameworks.

Key takeaways include:

- Data privacy, bias mitigation, and transparency are primary concerns that require further regulatory clarity.
- Institutional oversight mechanisms need reinforcement to prevent AI-related academic integrity violations.
- Current AI policies must evolve beyond general guidelines to include enforceable actions and accountability structures.
- Stronger collaboration between academia and the AI sector is essential to ensure ethical AI applications.

8.2 | Second Round Findings and Analysis

This section provides an evaluation of the second round of the Delphi study, where participants assessed their level of agreement on various aspects of AI ethics and governance through a Likert scale. The analysis incorporates interquartile range (IQR) values to measure the extent of consensus among respondents. Table 2 summarises the areas of consensus and divergence among experts, providing insight into the ethical dimensions most in need of clearer policy or guidance.

AI-driven feedback mechanisms remain a topic of debate regarding their alignment with ethical standards, as indicated by the level of divergence in expert responses (IQR = 1.0).

The second phase of the Delphi study offered critical insights into expert perspectives on AI ethics, governance, and academic integrity, highlighting both areas of agreement and points of contention. The analysis of Likert-scale responses, combined with interquartile range (IQR) values, provided a clearer understanding of the prevailing trends in expert opinions.

While certain aspects of AI ethics and regulatory frameworks received broad consensus, multiple areas still require further exploration and policy refinement. The results suggest that while ethical guidelines for AI research are in place, their interpretation and enforcement differ across institutions. This is

particularly evident in responses concerning transparency and accountability in AI-driven data usage, where moderate divergence (IQR = 1.0) indicates lingering uncertainty about the adequacy of current measures.

The most pronounced disagreement emerged in relation to ethical standards for data sharing (IQR = 2.0). Experts offered varying perspectives on whether clear ethical boundaries for data exchange exist; pointing to a lack of universally accepted regulations. Similarly, concerns surrounding unauthorised access to personal data (IQR = 1.5) indicate a pressing need for improved privacy protections and regulatory oversight. These concerns echo findings by Radanliev et al. (2024), who advocate for the adoption of privacy-preserving AI techniques, such as differential privacy, homomorphic encryption, and federated learning. Their work highlights how these tools can reinforce ethical AI deployment by embedding privacy safeguards at the algorithmic level.

On the other hand, topics related to AI bias and student performance assessment showed relatively strong consensus. Responses suggested a general agreement (IQR ≤ 0.5) that AI-driven academic performance models adhere to ethical principles, though some concerns about algorithmic bias persist. This suggests that while AI is largely seen as a fair tool in student evaluations, ongoing scrutiny is necessary to uphold integrity and transparency in assessment practices. Additionally, the discussion around how learners engage with generative AI technologies reveals further ethical complexities. Lodge et al. (2023) argue that the simplistic analogy of generative AI as a calculator is insufficient. Their typology of learner-AI interactions introduces the concepts of cognitive offloading, mind extension, and co-regulated learning, suggesting that the ethical dimensions of AI in education are inseparable from how students cognitively and socially engage with these tools. This nuanced view emphasises that ethical guidelines must evolve to accommodate the various ways students are integrating AI into their learning processes, intentionally or unintentionally, thus demanding policies that consider learner agency and context-driven use.

Another key theme that emerged from the findings is the role of institutional policies and funding structures in AI ethics enforcement. Responses displayed moderate divergence (IQR = 1.0) regarding the effectiveness of sanctions for ethical violations, implying that current enforcement mechanisms may require reinforcement. Similarly, uncertainty remains over the extent to which funding agencies actively promote ethical AI research, underscoring the need for clearer accountability measures. Radanliev (2025) underscores the role of international AI policy frameworks in shaping these institutional norms. His comparative study across EU, USA and Chinese frameworks illustrates the challenges of harmonising transparency, fairness, and privacy across jurisdictions. He recommends strategies such as institutional audits, fairness-aware development processes, and diverse stakeholder inclusion as means to achieve ethical compliance. Furthermore, Xiao et al. (2025) expose deep-seated uncertainties around the responsible use of generative AI in education. Their exploration of the ‘grey areas’—particularly those concerning authorship, autonomy, data use, and assessment integrity—reinforces this study’s findings that ethical ambiguities persist, especially where institutional clarity is lacking. They call for critical reflection on what constitutes

TABLE 2 | Summary of consensus levels on ethical AI statements in higher education (Delphi Round 2).

Question	IQR value	Interpretation
Ethical guidelines in AI research are clearly defined	0.5	High agreement on the clarity of ethical guidelines
Transparency and accountability in data usage are ensured	1.0	Moderate disagreement on the enforcement of transparency
Mechanisms exist to uphold academic integrity in AI research	1.0	Mixed opinions on the effectiveness of academic integrity measures
AI systems operate without causing bias	0.5	General agreement that AI bias remains an issue
AI decisions undergo ethical review	0.5	Most respondents agree that AI decisions are ethically reviewed
Unauthorised use of personal data is prevented	1.5	Significant concerns over the unauthorised use of personal data.
Data anonymization policies are sufficient	1.0	Varied opinions on the adequacy of data anonymization policies
Clear ethical boundaries exist for data sharing	2.0	Largest disagreement, indicating a lack of consensus on ethical data sharing.
Data collection processes in academic research are reliable	1.0	Moderate concerns about the reliability of data collection processes
Existing AI policies are comprehensive and sufficient	1.0	Some disagreement on whether AI policies are sufficient
Guidelines supporting academic integrity are effectively enforced	1.0	Uncertainty about how well integrity guidelines are enforced
Sanctions for ethical violations are effective deterrents	1.0	Sanctions for ethics violations may not be effective enough
Ethical AI standards are widely adopted	1.0	Ethical AI adoption is inconsistent across institutions
Collaborations between academia and AI industry comply with ethical standards	1.0	Uncertainty about the ethical alignment of academia-industry collaborations
Ethical guidance mechanisms exist in AI projects	1.0	Some respondents believe AI projects have ethical guidance, others disagree
Industry-backed academic research maintains independence	1.0	Concerns about corporate influence on academic independence
Funding bodies actively promote ethical standards	1.0	Unclear if funding bodies effectively promote AI ethics
AI is used impartially in student assessments	0.5	General agreement on the impartiality of AI in assessments.
AI-driven academic performance models adhere to ethical principles	0.0	Strong consensus that academic performance models follow ethical principles
AI fairly evaluates student performance	0.5	Minor concerns about fairness in AI-based student evaluations

meaningful, fair, and educative use of AI, warning that policy vacuums may lead to misaligned practices. Their collective call for a more interdisciplinary and educator-centred approach to AI policy echoes the Delphi panel's view that stakeholder voices must shape ethical guidelines from the ground up. These themes are further reinforced by Bozkurt et al. (2024), which offers a comprehensive, collective perspective on navigating the future of GenAI in education. The manifesto identifies GenAI as a transformative yet deeply contested force, emphasising that

it is not ideologically neutral and can amplify biases and inequities if left unchecked. The authors advocate for careful, inclusive, and critical integration of GenAI, underscoring the need for participatory policy design, ethical oversight, and preservation of human-centred learning values. Their call for evidence-informed, human-augmenting AI practices directly aligns with this Delphi study's findings on institutional responsibility, ethical ambiguity, and the importance of inclusive stakeholder input in educational AI governance.

The intersection of academia and the AI industry remains an area of mixed perspectives. While some experts believe that industry-sponsored academic research maintains independence, others raise concerns about potential conflicts of interest. This suggests a growing need for structured oversight to ensure that commercial influence does not compromise academic integrity.

In summary, the findings from this phase highlight critical areas where AI ethics and governance frameworks need improvement. Future research should prioritise closing gaps in data privacy regulations, strengthening institutional enforcement mechanisms and developing standardised ethical guidelines for AI integration in academia. As AI technology continues to advance, fostering an ongoing and transparent dialogue on these ethical considerations will be essential to ensuring responsible AI implementation in higher education and research.

9 | Interpretation of High Variability Questions

The elevated IQR values in these responses indicate a significant lack of consensus regarding data privacy measures and ethical regulations for AI applications. Several factors may contribute to this variation:

- **Different Interpretations of Data Protection Policies:** Participants may perceive existing regulations differently, leading to diverse assessments of their effectiveness (Roth et al. 2017)
- **Institutional and Regional Variations:** AI Governance Frameworks Differ Across Institutions and Countries, Shaping Respondents' Perspectives (Okoli and Pawlowski 2004)
- **Differences in Awareness Levels:** Individuals with greater exposure to AI ethics discussions may hold different expectations than those less familiar with the topic (Skulmoski et al. 2007)

Open-Ended Responses Provided Additional Context for These Discrepancies. One Participant Remarkd, 'Data protection policies vary widely across institutions, making it challenging to establish universally accepted ethical guidelines.' Another Respondent Observed, 'While some organizations enforce strict AI ethics regulations, others lack clear policies, leading to inconsistencies in implementation'.

These qualitative insights further validate the statistical variability observed; highlighting the fragmented landscape of AI ethics governance.

The findings are consistent with prior research on AI governance and ethics. Mittelstadt et al. (2016) identify similar challenges in defining clear ethical guidelines for AI applications, particularly within academic settings. Additionally, Rowe and Wright (1999) argue that agreement on ethical standards often depends on the maturity of regulatory frameworks and institutional priorities. Compared to these studies, our results suggest

that AI ethics policies continue to be highly fragmented, highlighting the need for more standardised approaches.

Although traditional Delphi studies frequently extend to three or more rounds, this study was intentionally concluded after two rounds, following best practices and practical considerations. Research indicates that when strong consensus is reached within the first two rounds, additional iterations contribute little to the findings, often resulting in redundancy rather than meaningful new insights (Okoli and Pawlowski 2004; Skulmoski et al. 2007; Roth et al. 2017).

While two key questions did not achieve full consensus, these unresolved issues offer valuable perspectives on ongoing ethical dilemmas rather than justifying the need for further rounds. Instead of excluding them from analysis, this study recognises these disagreements as significant discussion points, underscoring the importance of stronger policy frameworks rather than forcing premature agreement where divergence still exists (Hsu and Sandford 2007; Rowe and Wright 1999). This is further illustrated in Figure 1, which summarises key ethical concerns, institutional requirements, and policy directions necessary for ethical governance in AI research.

10 | Implications for AI Ethics and Policy

The findings highlight the pressing need for more clearly defined and widely recognised standards in AI ethics, with a particular focus on data privacy. The variation in expert perspectives suggests that several key aspects require attention. One crucial consideration is the establishment of standardised AI ethics frameworks. Institutions should develop transparent and consistent policies that provide a shared understanding of ethical boundaries, ensuring that all stakeholders operate within a well-defined set of guidelines (Dalkey and Helmer 1963).

In addition to policy development, increasing education and awareness around AI ethics is essential. Implementing targeted training programmes for AI researchers and practitioners can help bridge existing knowledge gaps and encourage greater alignment in ethical considerations (Turoff and Linstone 2002). By fostering a culture of ethical awareness, institutions can better equip individuals to navigate complex ethical dilemmas associated with AI applications.

However, policies and education alone are not sufficient; there must also be effective enforcement mechanisms in place to ensure compliance with ethical guidelines. Without proper oversight and accountability, even the most well-intentioned policies risk being overlooked or inconsistently applied. Establishing robust enforcement structures will be critical in maintaining adherence to ethical standards and ensuring that AI technologies are developed and used responsibly (Gordon 1994).

Recent developments in generative AI—such as ChatGPT, Claude and GitHub Copilot—have brought forward a new wave of ethical considerations in academic research. These tools, while enhancing productivity, have raised concerns regarding the authenticity of scholarly outputs. There is an increasing risk that such

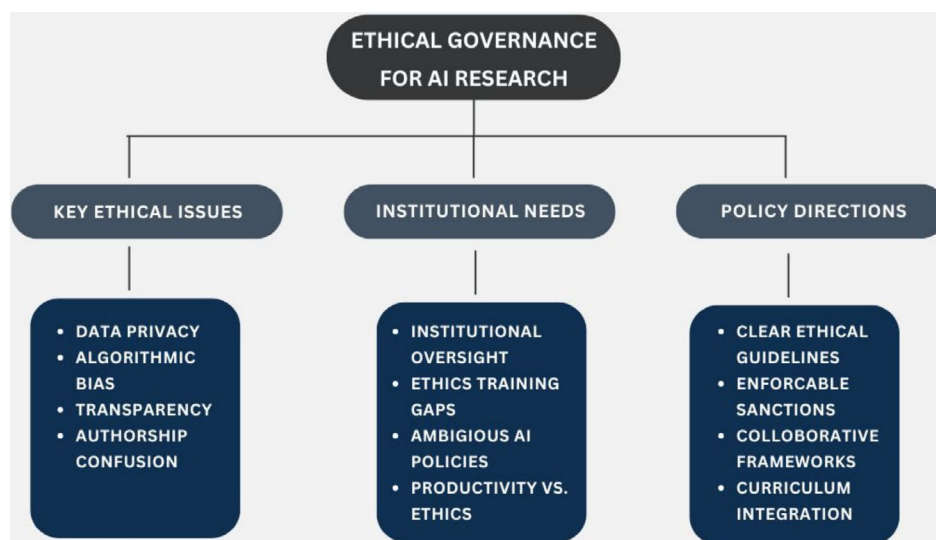


FIGURE 1 | Summary of key themes and recommendations.

technologies may be used to produce content that is indistinguishable from human work, making authorship and accountability more difficult to trace. Moreover, the datasets on which these systems are trained often remain opaque, leading to uncertainty about the inclusion of biased, copyrighted, or unethically sourced material. This situation presents a challenge for academic institutions, as traditional integrity frameworks may not adequately address these novel risks. It has therefore been suggested that clearer institutional policies and more robust AI ethics education should be implemented to ensure responsible and transparent use of generative AI in research and learning environments.

11 | Future Research Directions

Future research should expand on these findings by investigating key areas that could contribute to the development of more comprehensive AI ethics frameworks. One important direction is the exploration of AI ethics within various cultural and institutional settings. Given that ethical standards and regulatory practices differ significantly across regions, adopting a cross-cultural perspective may help in formulating globally relevant guidelines that accommodate diverse societal values and governance models. Furthermore, conducting longitudinal studies is essential for understanding how AI ethics policies evolve over time. By examining the implementation and ongoing adjustments of these policies, researchers can assess their effectiveness and pinpoint areas that require improvement. Such investigations would provide deeper insights into how institutions navigate emerging ethical dilemmas and adapt to shifts in regulatory landscapes. Another valuable approach involves utilising mixed-method research to analyse AI ethics in practical contexts. Combining Delphi studies with case studies or ethnographic research can offer a more detailed understanding of the challenges faced by institutions and policymakers. By merging expert consensus with in-depth qualitative insights, future studies can bridge the gap between theoretical considerations and real-world applications, ultimately leading to more well-informed and actionable policy recommendations.

12 | Conclusion

In light of the urgent ethical challenges identified through this Delphi study, the education sector must undertake immediate and coordinated measures to adapt effectively to the evolving AI landscape. First, higher education institutions should establish comprehensive and enforceable AI ethics frameworks that provide clear guidance on issues such as authorship attribution, data governance, and the preservation of academic integrity. These frameworks must move beyond aspirational principles and translate into actionable institutional policies. Second, it is imperative to implement structured professional development initiatives aimed at enhancing the AI literacy of academic staff, researchers, and administrative personnel. Such initiatives should emphasise ethical reasoning, case-based learning, and the responsible integration of AI tools within pedagogical and research contexts. Third, sustained multi-stakeholder collaboration involving academic institutions, AI developers, legal experts, and policymakers is essential for the co-construction of dynamic governance models that remain responsive to technological advancements. These measures, if promptly enacted, will serve to bridge current regulatory gaps, foster a culture of ethical responsibility, and ensure that the integration of AI technologies within higher education aligns with the core values of academic excellence, integrity, and societal trust.

The integration of AI into higher education has brought about transformative opportunities, but it has also introduced complex ethical challenges that demand immediate and thoughtful action. This Delphi study, engaging a diverse panel of 12 international experts, has shed light on the critical ethical dilemmas and institutional barriers associated with AI-driven research, offering actionable recommendations to uphold academic integrity and ensure the responsible use of AI technologies. The findings underscore the pressing need for clear ethical guidelines, robust institutional oversight, and enhanced collaboration between academia and industry to navigate the evolving ethical landscape shaped by AI. The study identified data privacy, algorithmic bias, and transparency as the most pressing

ethical challenges in AI research. These issues highlight the importance of developing enforceable policies that protect personal data, mitigate bias, and ensure accountability in AI decision-making processes. Furthermore, the lack of institutional oversight and the prioritisation of productivity over ethics were identified as significant barriers to maintaining academic integrity. Addressing these challenges requires a fundamental shift in institutional priorities, with a greater emphasis on ethical responsibility and the establishment of stronger governance mechanisms. The findings also revealed that while current AI policies provide a foundational level of ethical boundaries, they often fall short in terms of enforceability and specificity. To bridge this gap, institutions must transition from broad principles to concrete, actionable guidelines that can be effectively implemented across diverse academic settings. Additionally, fostering collaboration between academia and the AI industry is essential to ensure that ethical considerations are integrated into AI research partnerships from the outset. In conclusion, this study emphasises the need for a flexible and evolving approach to AI ethics in higher education. As AI technologies continue to advance, the ethical frameworks that guide their development and application must also evolve. By fostering ongoing collaboration among academia, policymakers, and industry, and by prioritising transparency, accountability, and inclusivity, institutions can uphold academic integrity and ensure that AI is used responsibly to advance knowledge and innovation. The findings of this study serve as a critical step toward creating a sustainable and ethically sound AI ecosystem in academia, one that balances the transformative potential of AI with the imperative to protect ethical values and public trust.

Author Contributions

Both authors contributed equally and collaboratively to all stages of the research. Specifically, they jointly designed and refined the Delphi study framework, selected and recruited expert participants, developed and piloted the Delphi questionnaires, conducted both rounds of data collection and analysis, and interpreted the results in line with the study's aims and ethical considerations. They also co-wrote and revised the manuscript. All authors have read and approved the final version of the manuscript.

Ethics Statement

This study was approved by the Anadolu University Ethics Committee (Approval No: 41/102, Date: 11/12/2024). All participants were informed about the aims and procedures of the research and provided their informed consent prior to participation.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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